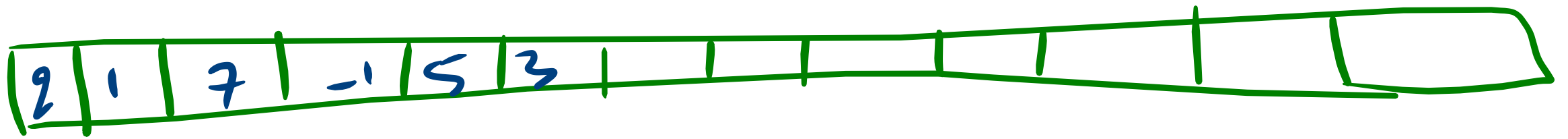


Welcome to this
Rakcha CP

X O X O ♡



0 1 2 . . . v.size() - 1

vector < vector < int > > v;

v.push_back(v[i]);

56. Merge Intervals

Solved 

Medium

Topics

Companies

Given an array of `intervals` where `intervals[i] = [starti, endi]` merge all overlapping intervals, and return an array of the non-overlapping intervals that cover all the intervals in the input.

$[1, 4], [4, 5]$

=

$[1, 5]$

Constraints:

- $1 \leq \text{intervals.length} \leq 10^4$
- $\text{intervals}[i].\text{length} == 2$
- $0 \leq \text{start}_i \leq \text{end}_i \leq 10^4$

$[2, 4]$, $[1, 6]$

$[1, 6]$

$[10, 15]$

$[1, \cancel{2}, \cancel{3}, 6, 8,$
 $10, 15, 18]$

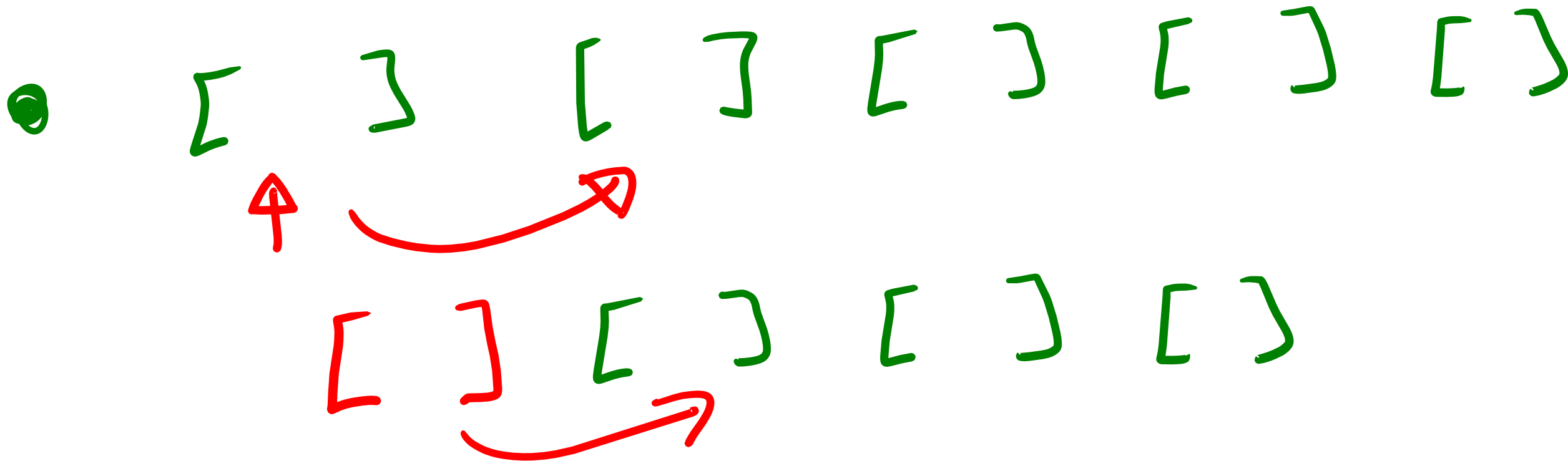
$[[1, 3], [2, 6], [8, 10], [15, 18]]$

$[1, 6] [8, 10] [15, 18]$

$[1, 2]$ $[2, 3]$
↑ ↑

Fedi

• Intervals Should be Sorted



Fedi

$[b_i, e_i]$

Youssef

$b_i < b_j$

$b_i = b_j$

$[b_j, e_j]$

$[1, 4] \quad [1, 3]$

$[1, 3] \quad [1, 4]$

$\rightarrow e_i < e_j$

• $[b_i, e_i]$ $[b_j, e_j]$

How to merge these
two intervals?

$[1, 7]$ $[2, 3]$

given: $b_i < b_j$

or $b_i == b_j$ &&

$e_i < e_j$

Result: $[b_i, \max(e_i, e_j)]$

$[1, 4] \times [5, 6]$
 $b_i \quad e_i \quad b_j \quad e_j$
 $[4, 5]$

$$e_i > b_j$$

$$4 > 5 \quad \times$$

$[1, 4] \quad [5, 6]$
 $\times$

given
 sorted
 intervals



$$[2, 4] \overset{[4, 6]}{\times} [6, 7]$$

$b-i \quad e-i \qquad b-j \quad e-j$

$$\boxed{b-j \leq e-i}$$

$6 \quad \cancel{4}$

$$[2, 6] \quad [6, 7] \xrightarrow{6 \leq 6 \checkmark} [2, 7]$$

$$[2, 4] \quad [3, 7] \longrightarrow [2, 7]$$

$$\underline{3 \leq 4 \checkmark}$$

Ayoub

$[1, 3]$

$[4, 6]$

$[2, 8]$

$[1, \cancel{2}, \cancel{3}, \cancel{4}, \cancel{6}, 8]$

$2 \cdot 10^4$

$[1, 8]$

$\downarrow$
 10^4

$[1, 3, 5]$

$[1, 3] [3, 5]$

$2 \cdot 10^8$